MTH642 - Fluid Mechanics (Quiz No. 4)

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Time Left 73 sec(s)

MTH642 - Fluid Mechanics (Quiz No. 4)

Quiz Start Time: 12:33 PM, 07 July 2025

Question # 10 of 10 (Start time: 12:36:43 PM, 07 July 2025) **Total Marks:** 1

Which of the following is a polar form of $\vec{\nabla}.(\rho\vec{V}) = 0$? Where $\vec{V} = (u_r, u_\theta, u_z)$.

Select the correct option

Reload Math Equations

- $\frac{1}{r}\frac{\partial(\rho r u_r)}{\partial r} + \frac{\partial(\rho r u_\theta)}{\partial \theta} + \frac{\partial(\rho u_z)}{\partial z} = 0$
- $\frac{1}{r}\frac{\partial(\rho r u_r)}{\partial r} + \frac{1}{r}\frac{\partial(\rho r u_\theta)}{\partial \theta} + \frac{\partial(\rho u_z)}{\partial z} = 0$
- $\frac{1}{r}\frac{\partial(\rho r u_r)}{\partial r} + \frac{\partial(\rho u_\theta)}{\partial \theta} + \frac{\partial(\rho u_z)}{\partial z} = 0$
- $\frac{1}{r}\frac{\partial(\rho r u_r)}{\partial r} + \frac{1}{r}\frac{\partial(\rho r u_\theta)}{\partial \theta} + \frac{\partial(\rho r u_z)}{\partial z} = 0$

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Quiz Start Time: 12:33 PM, 07 July 2025

Question # 9 of 10 (Start time: 12:36:24 PM, 07 July 2025) **Total Marks:** 1

The Froude number (Fr) is equal to _____.

Select the correct option

Reload Math Equations

- $Fr = \frac{V}{gL}$
- $Fr = \frac{\sqrt{gL}}{V}$
- $Fr = \frac{V}{\sqrt{gL}}$
- $Fr = \frac{gL}{V}$

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Quiz Start Time: 12:33 PM, 07 July 2025

Question # 8 of 10 (Start time: 12:36:05 PM, 07 July 2025) **Total Marks:** 1

For the unsteady compressible three dimensional flow, the continuity equation is given as _____.

Select the correct option

Reload Math Equations

- (●) $\partial\rho/\partial t + \partial(\rho u)/\partial x + \partial(\rho v)/\partial y + \partial(\rho w)/\partial z = 0$
- ( ) $\partial\rho/\partial t + \rho\partial(u)/\partial x + \rho\partial(\rho v)/\partial y + \rho\partial(u)/\partial z = 0$
- ( ) $\partial u/\partial x + \partial v/\partial y + \partial w/\partial z = 0$
- ( ) $\partial(\rho u)/\partial x + \partial(\rho v)/\partial y + \partial(\rho u)/\partial z = 0$

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Quiz Start Time: 12:33 PM, 07 July 2025

Question # 7 of 10 (Start time: 12:35:33 PM, 07 July 2025) **Total Marks:** 1

_____ is the incompressible continuity equation in Cylindrical coordinates.

Select the correct option

Reload Math Equations

- ( ) $\frac{1}{r}\frac{\partial(ru_r)}{\partial r} + \frac{\partial(ru_\theta)}{\partial\theta} + \frac{\partial(ru_z)}{\partial z} = 0$
- (●) $\frac{1}{r}\frac{\partial(ru_r)}{\partial r} + \frac{1}{r}\frac{\partial(ru_\theta)}{\partial\theta} + \frac{\partial(ru_z)}{\partial z} = 0$
- ( ) $\frac{1}{r}\frac{\partial(ru_r)}{\partial r} + \frac{1}{r}\frac{\partial(ru_\theta)}{\partial\theta} + \frac{\partial u_z}{\partial z} = 0$
- ( ) $\frac{1}{r}\frac{\partial(ru_r)}{\partial r} + \frac{\partial u_\theta}{\partial\theta} + \frac{\partial u_z}{\partial z} = 0$

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Quiz Start Time: 12:33 PM, 07 July 2025

Question # 6 of 10 (Start time: 12:35:10 PM, 07 July 2025) **Total Marks:** 1

For the steady incompressible two dimensional flow, the continuity equation is given as _____.

Select the correct option

Reload Math Equations

- $\partial\rho/\partial t + \partial(\rho u)/\partial x + \partial(\rho v)/\partial y = 0$
- $\partial u/\partial x + \partial v/\partial y = 0$
- $\partial\rho/\partial t + \partial u/\partial x + \partial v/\partial y = 0$
- $\partial(\rho u)/\partial x + \partial(\rho v)/\partial y = 0$

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MTH642 - Fluid Mechanics (Quiz No. 4)

Quiz Start Time: 12:33 PM, 07 July 2025

Question # 5 of 10 (Start time: 12:34:52 PM, 07 July 2025) **Total Marks:** 1

Kilogram is the unit used to measure the dimension of _____.

Select the correct option

Reload Math Equations

- amount of matter
- amount of light
- time
- mass

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Quiz Start Time: 12:33 PM, 07 July 2025

Question # 4 of 10 (Start time: 12:34:32 PM, 07 July 2025) **Total Marks:** 1

Which of the following is a general form of continuity equation?

Select the correct option

Reload Math Equations

- $\vec{\nabla}.(\rho\vec{V}) = 0$
- $\frac{\partial\rho}{\partial t} + \vec{\nabla}.(\rho\vec{V}) = 0$
- $\frac{\partial\rho}{\partial t} + \rho\vec{\nabla}.\vec{V} = 0$
- $\frac{\partial\rho}{\partial t} + \vec{V}.(\rho\vec{\nabla}) = 0$

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Quiz Start Time: 12:33 PM, 07 July 2025

Question # 3 of 10 (Start time: 12:34:06 PM, 07 July 2025) **Total Marks:** 1

The dimension of the Bernoulli constant is _____.

Select the correct option

Reload Math Equations

- mL^2t^{-2}
- $mL^{-1}t^{-2}$
- $mL^{-1}t^{-1}$
- $mL^{-2}t^{-2}$

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Quiz Start Time: 12:33 PM, 07 July 2025

Question # 2 of 10 (Start time: 12:33:45 PM, 07 July 2025) **Total Marks:** 1

The flow is appoximated as incompressible if _____.

Select the correct option

Reload Math Equations

- $\nabla \cdot V = \infty$
- $\nabla \cdot V = 0$
- $\nabla \cdot V = -1$
- $\nabla \cdot V = 1$

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MTH642 - Fluid Mechanics (Quiz No. 4)

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Quiz Start Time: 12:33 PM, 07 July 2025

Question # 1 of 10 (Start time: 12:33:21 PM, 07 July 2025) **Total Marks:** 1

Kilogram is the unit used to measure the dimension of _____.

Select the correct option

Reload Math Equations

- amount of matter
- time
- mass
- amount of light

Click to Save Answer & Move to Next Question

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